

The War of the Currents:

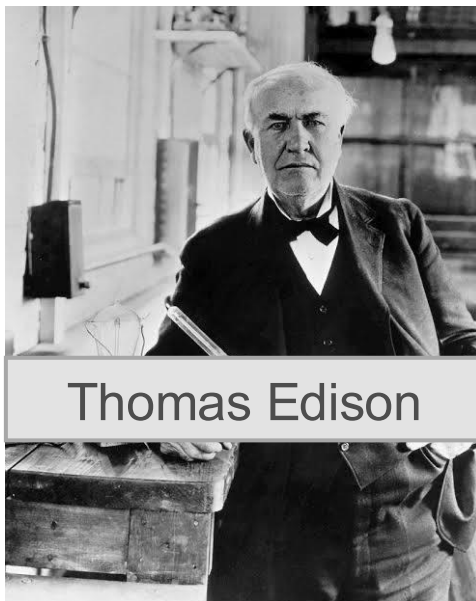


A REAL Scientific Debate



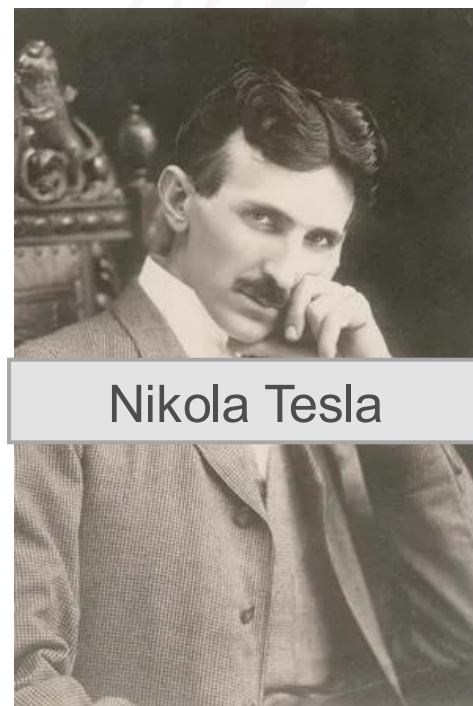
The Current Wars: AC vs DC

- **Edison** bet on **DC** while **Westinghouse** and **Tesla** bet on **AC**
- **AC had a physics advantage:** it can be easily "stepped up" to high voltage (lower resistive losses) and sent long distances then "stepped down" to your house
- That's why Niagara could light up Buffalo, and eventually your phone charger



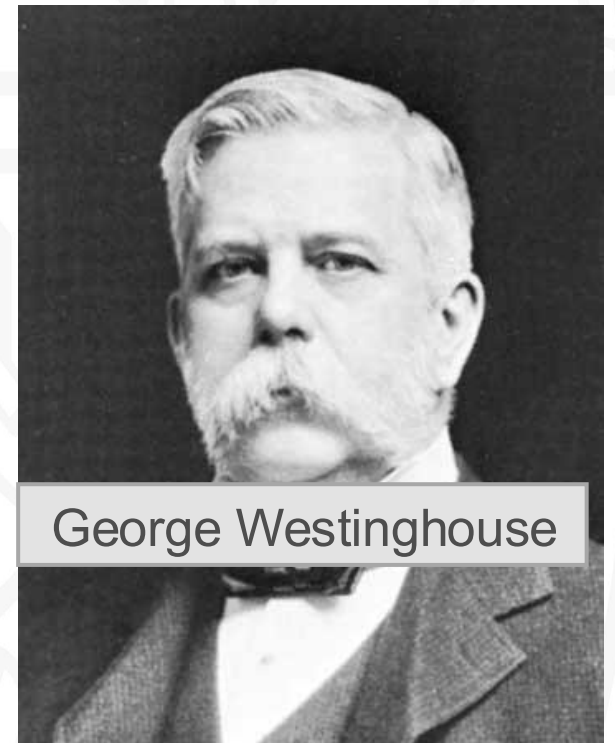
Thomas Edison

<https://www.britannica.com/biography/Thomas-Edison>



Nikola Tesla

<https://www.britannica.com/biography/Nikola-Tesla>



George Westinghouse

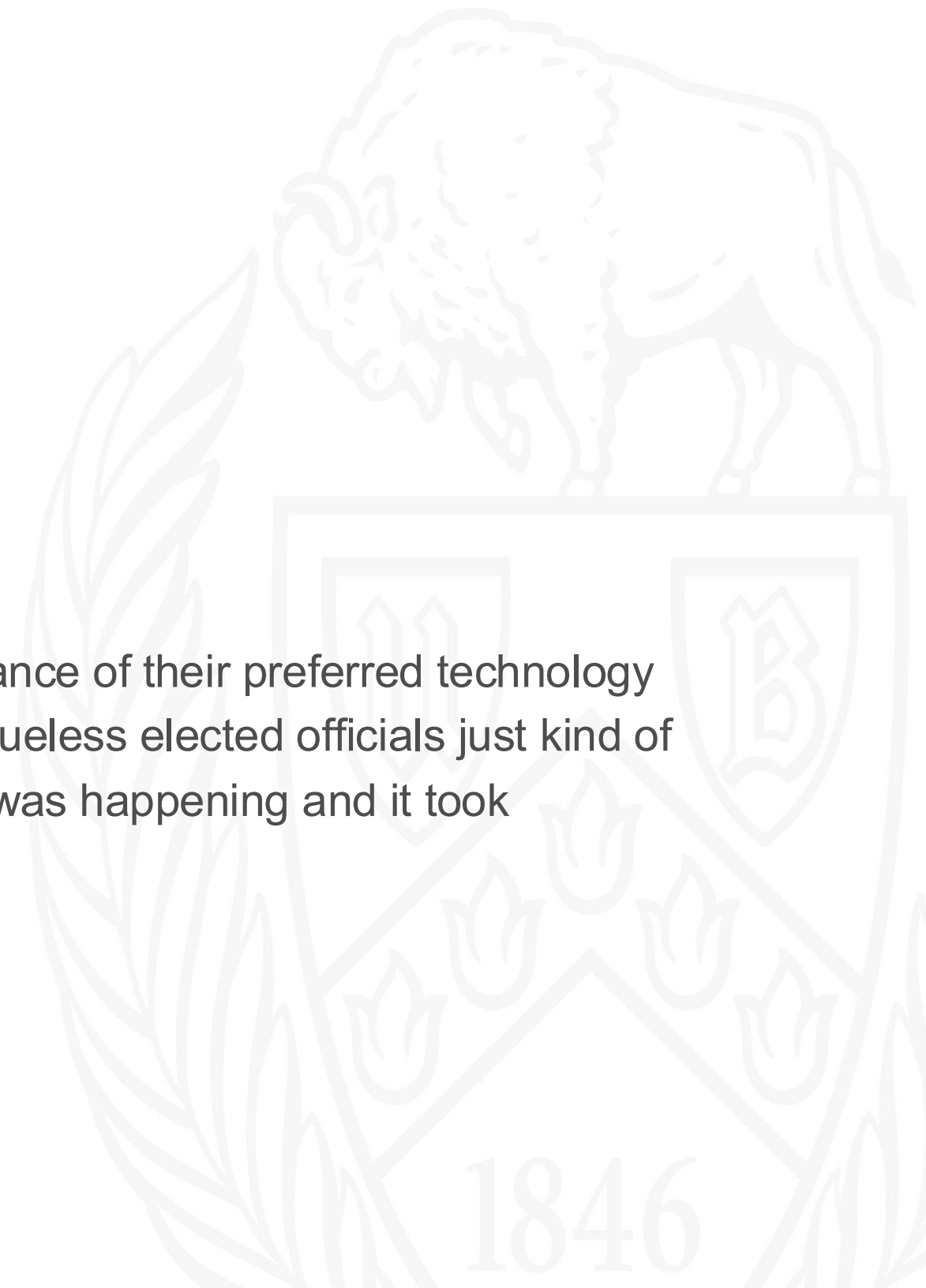
<https://www.britannica.com/biography/George-Westinghouse>

AI used to create some of these slides

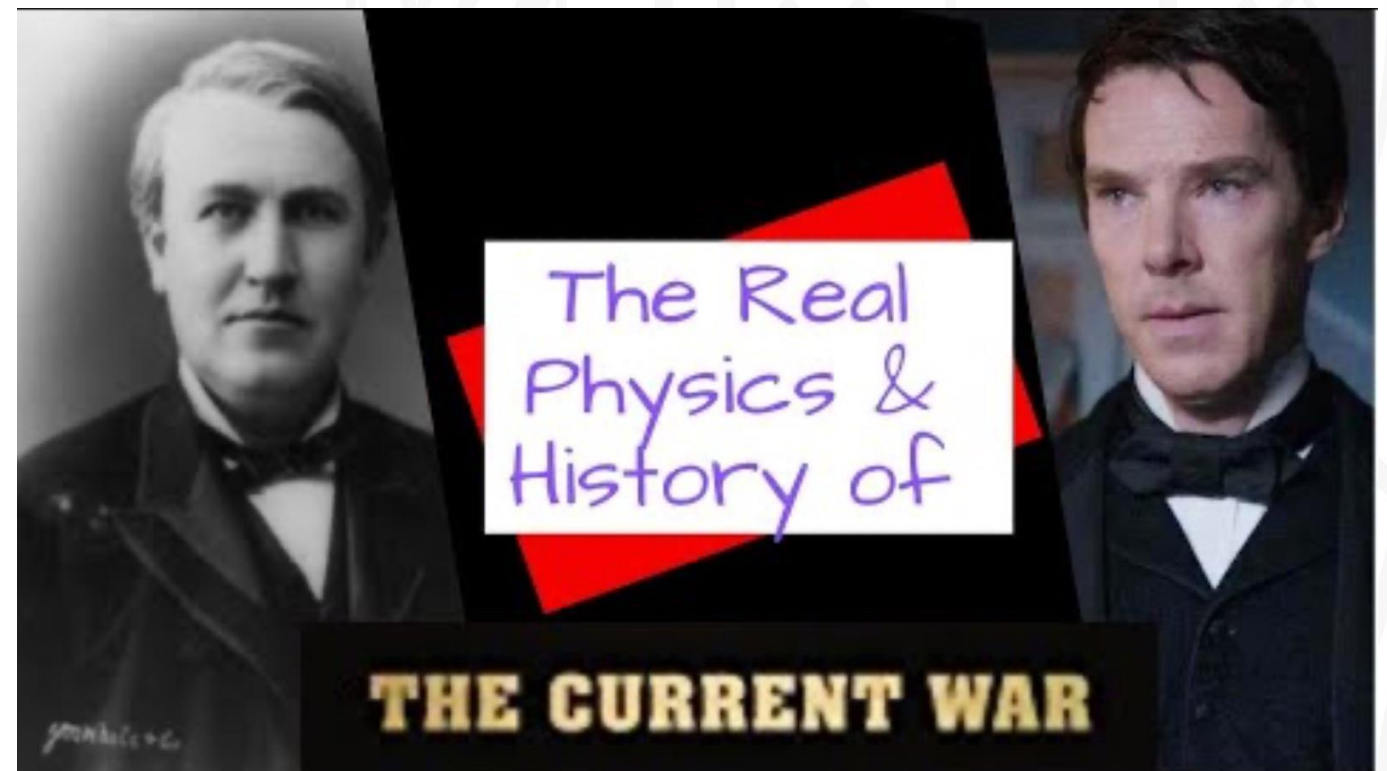
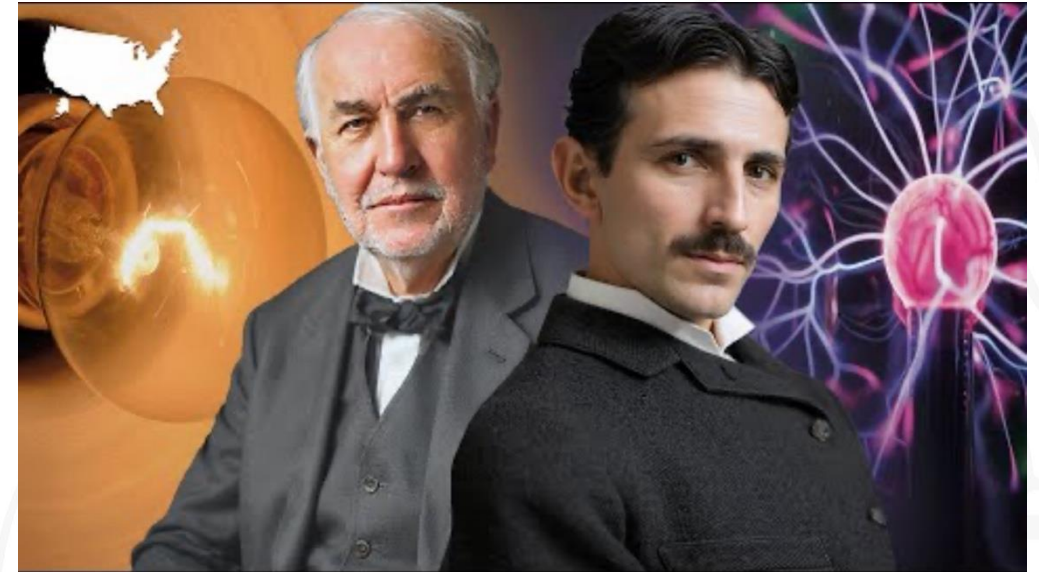
- I am going to be showing you how to effectively use AI during this class
- For this set of slides, I had Claude summarize the Wikipedia article on the [War of the Currents](#) to create a timeline and make some biography slides.
- You're allowed to use AI for this class, but you must cite a few things:
 - Tokens processed (for this, 13.1M)
 - Estimated actual cost (for this, around \$20)
 - Power used (for this, around 0.3 kWh, running a microwave for 15-40 minutes)
 - Water used (for this, around 0.5 L, around a glass of water)

Content warnings

- The current wars were MESSS-SYYYY
- Contains:
 - Death by electrocution
 - Animal cruelty and euthenasia
 - Collusion
 - Business oligarchs fighting to attain national dominance of their preferred technology regardless of the cost to humanity while relatively clueless elected officials just kind of let them do it because they didn't understand what was happening and it took decades to sort out to be safe for humans
 - I can't imagine this happening again...



Social Media



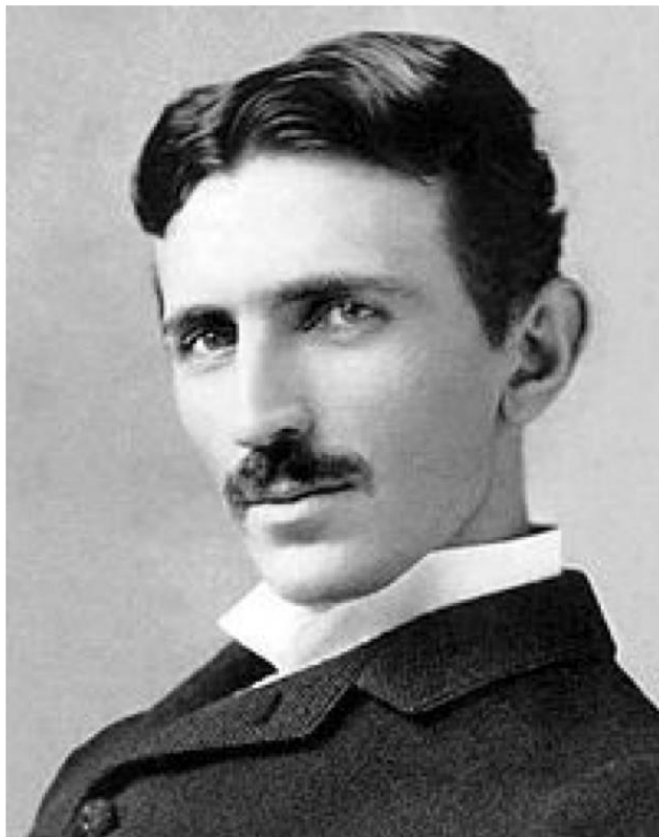
Pop Culture Weighs In

- [The Current War](#): Pretty good film in 2017!
- Got the gist of the debate and history right and only a few “artistic license” changes

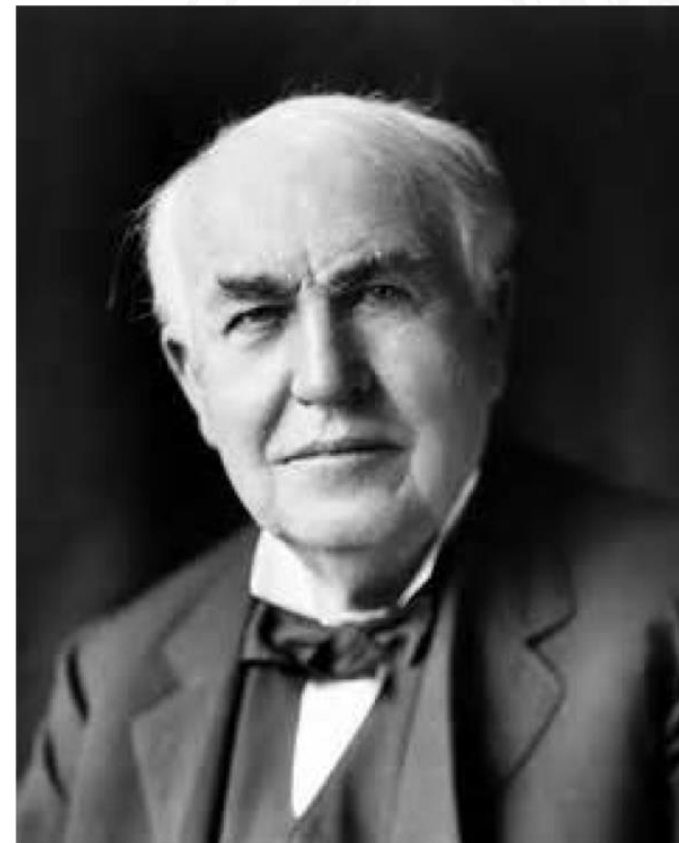


AC versus DC Power : A REAL Scientific Debate

- Tesla (with Westinghouse's backing) Versus Edison



Foreigner, well educated,
eccentric “societal elite”

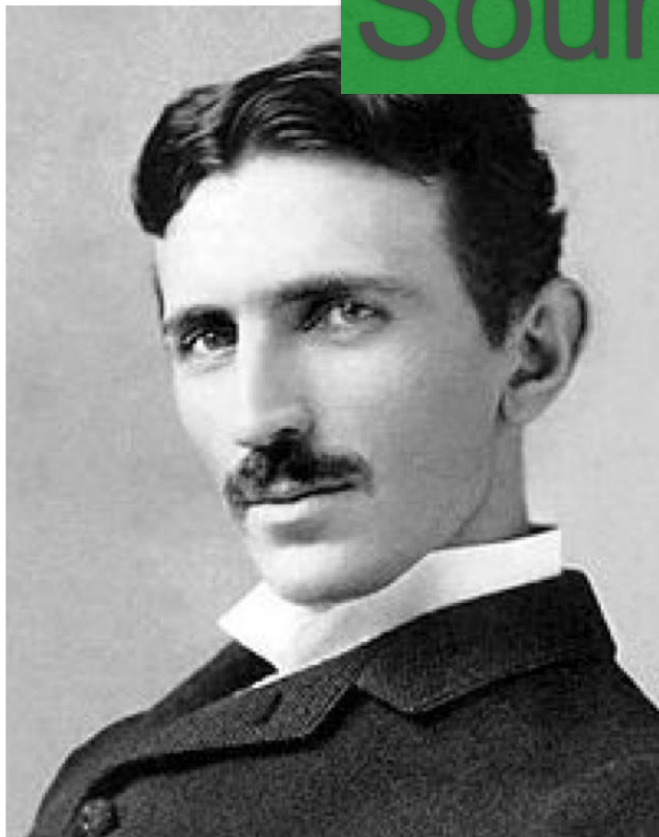


US born, no formal
education, “common person”⁷

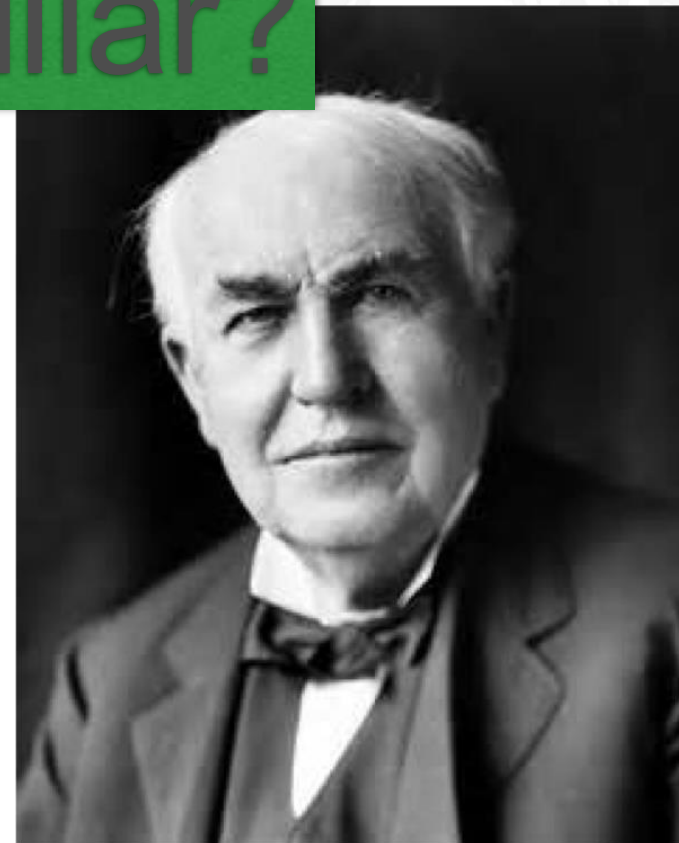
AC versus DC Power : A REAL Scientific Debate

- Tesla (with Westinghouse's backing) Versus Edison

Sound familiar?



Foreigner, well educated,
eccentric “societal elite”



US born, no formal
education, “common person”⁸

AC versus DC Power : A REAL Scientific Debate

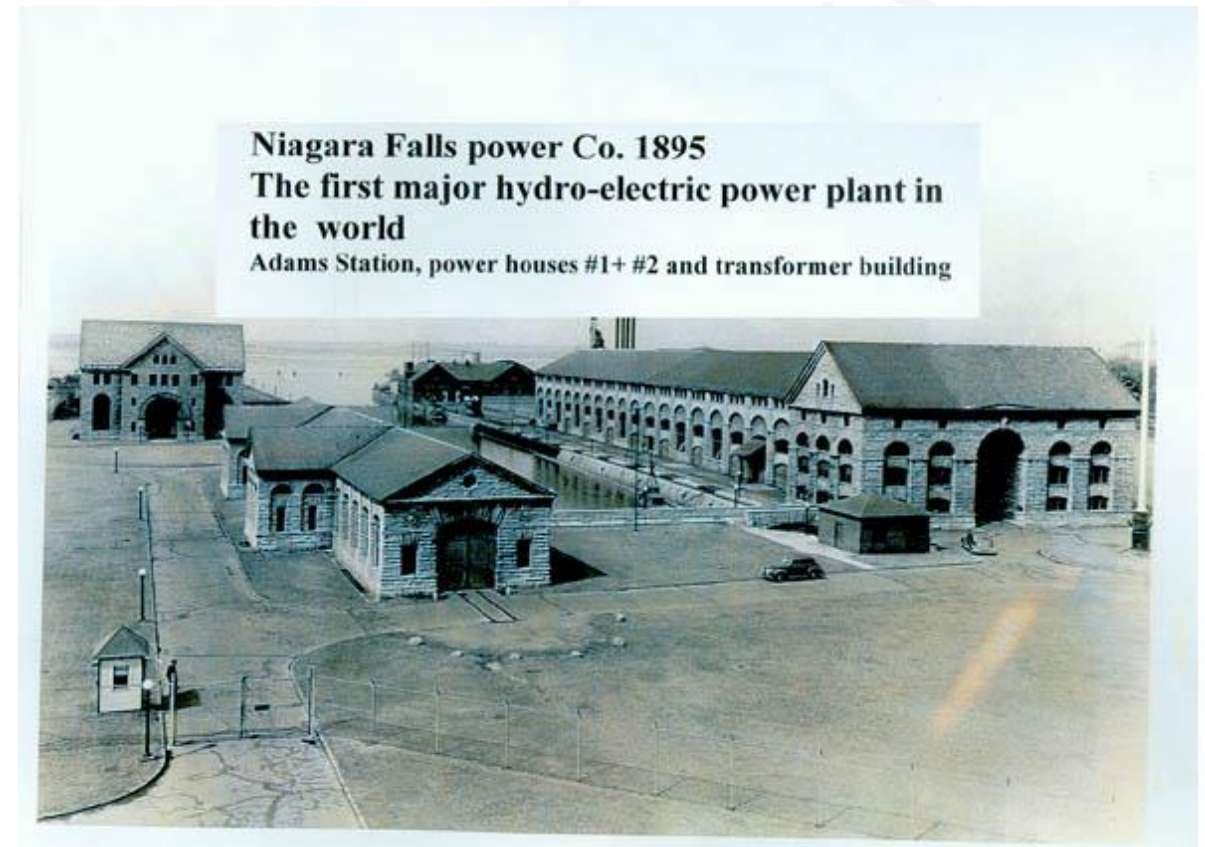
	AC Power	DC Power
Pros	- Generated naturally by turbines - Efficient energy transfer over long distances	- More efficient (RMS power versus constant power) - Recently: generated by solar panels
Cons	- Uses high voltage power lines for energy transmission	- Huge energy losses over long distances

AC versus DC Power : A REAL Scientific Debate

- The politics:
 - Eccentric “elite” foreigner versus home-town boy
 - Political machinery from Edison (“**special interest groups!**”)
 - Lobbying local and national governments
 - AC was dangerous due to high voltage power lines
 - AC was used for the electric chair
 - AC power has “inferior quality” (uhhhh.... no)
 - Good old fashioned corruption and collusion!
- The economics:
 - Westinghouse, Edison Electric Light Company, Thomas-Houston Electric Company : three huge companies fighting for control of a giant emerging market
 - Westinghouse and Thomas-Houston used AC
 - Edison Electric used DC
 - All out economic, social, and political war for a decade

AC versus DC Power : A REAL Scientific Debate

- Resolution came BOTH from the economics and the science :
 - DC required too many generators
 - Turbines naturally generate AC power from Faraday's Law
 - AC power was able to supply electricity to remote areas
 - In the end, Tesla's ideas won out
- Eventually the big companies for AC and DC merged together and adopted AC. Everyone involved walked away much richer than before and we haven't heard about it since.



Electric Power started here — and it was clean

- The [Adams Power Plant at Niagara Falls \(1895\)](#) was the first **large-scale AC (hydro!)electric** system in history
- The very first big power project humanity built was **renewable**
- It also stole land from the Haudenosaunee people without compensation, in violation of the 1794 Treaty of Canandaigua. (oh wow I'm sooooo surprised we did that... again... [oh and again in the 1960's](#))
- It powered Buffalo and proved electricity could travel long distances



Robert Moses.
Hopefully now rotting in hell

<https://www.newyorkalmanack.com/2025/01/robert-mosess-niagara-st-lawrence/>

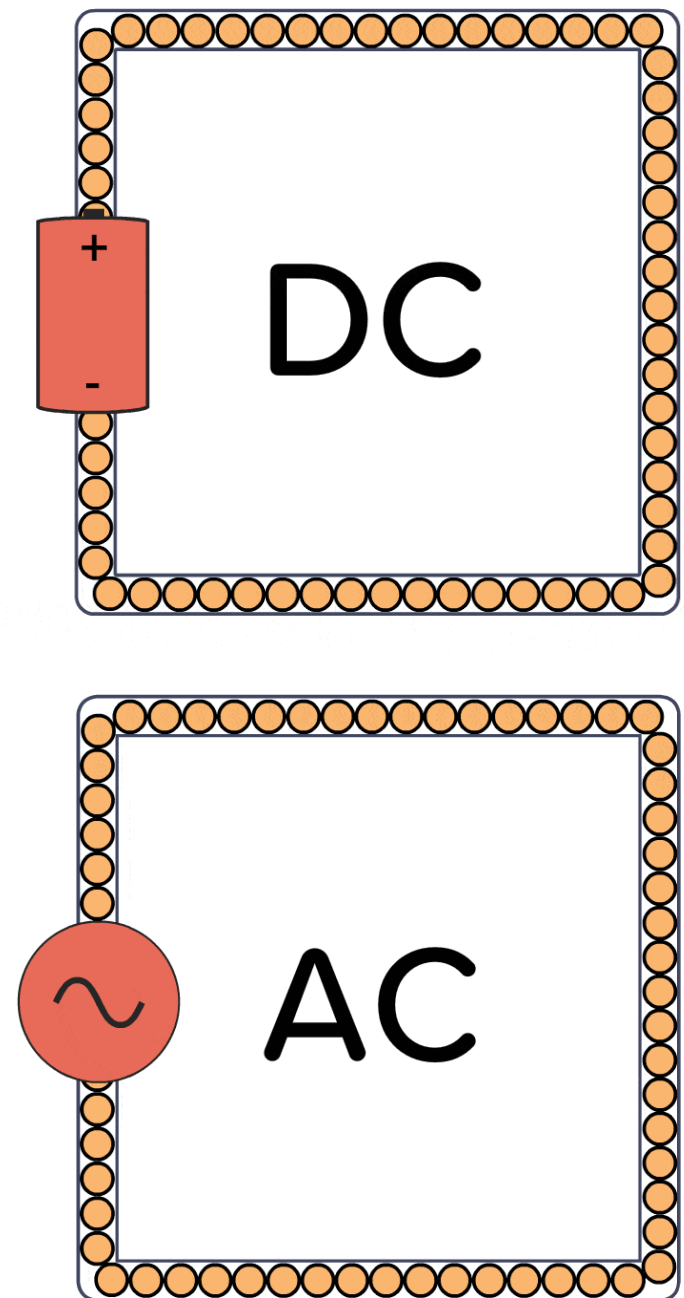
THE WAR OF THE CURRENTS

AC vs. DC, 1882–1896



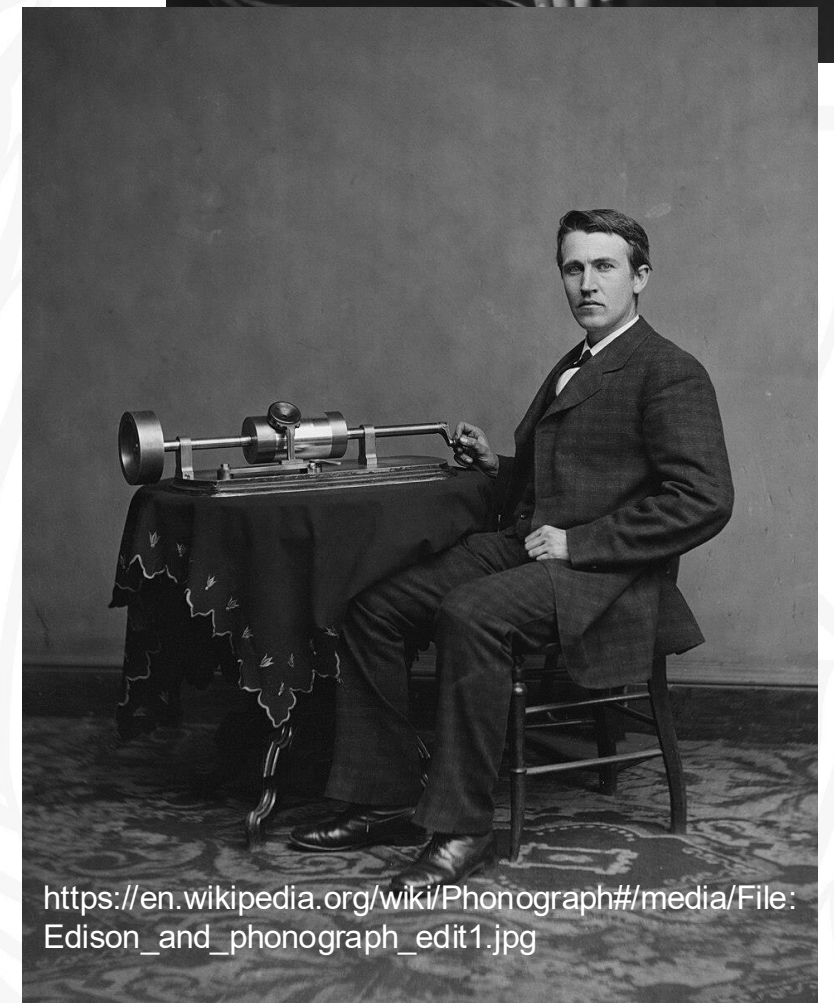
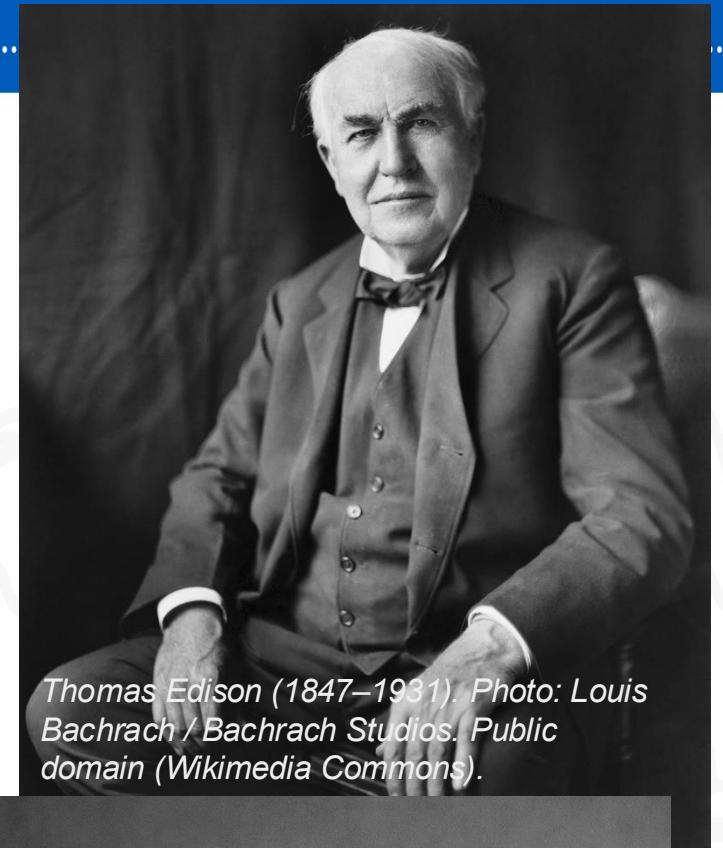
The War of the Currents: What Were They Fighting Over?

- AC versus DC current:
 - DC (Edison): flows one way, like a battery. Hard to change voltage, so it dies within ~1 mile of the plant.
 - AC (Westinghouse/Tesla): reverses ~60x per second. A transformer steps it up to high voltage for cheap long-distance travel, then back down at your house.
- **Recall Lecture 2:** power lost in a wire is I^2R . High voltage = low current = low power loss over long distances



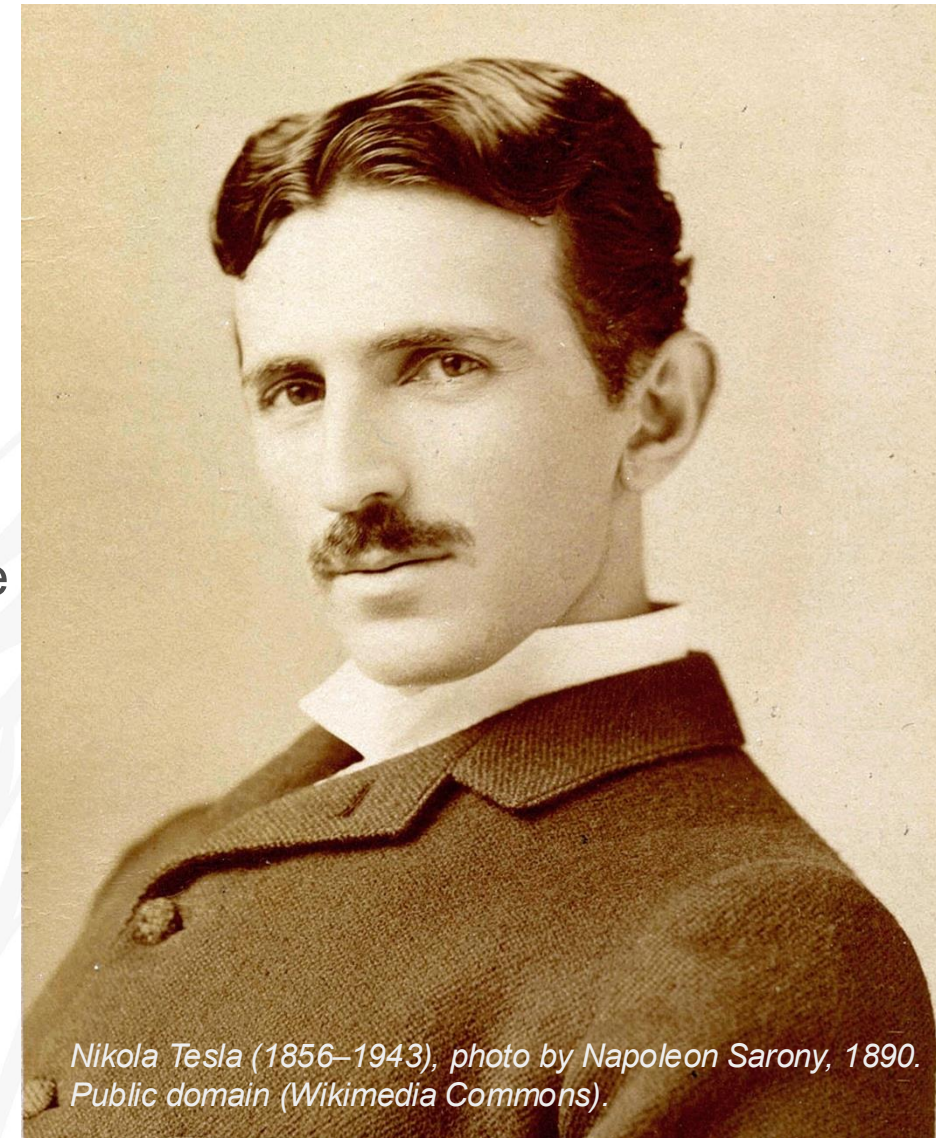
Thomas Edison: the Wizard of Menlo Park

- Born in Ohio 1847, raised in Michigan
- No formal education, worked as a railroad telegrapher, sold patents for improvements to telegraphs and moved to NYC to be an inventor
- Invented the [phonograph](#) (not the lightbulb), built the first DC grid and utility.
- Inventive genius but also a ruthless businessman who stole many ideas and just had better marketing (like the lightbulb and the radio)
 - Also kind of an a\$\$hole
- Backed DC because he owned the distribution. When the physics turned against him, he ran a fear campaign instead of adapting.
- Lost control of his company in the 1889 merger; walked away to iron-ore mining.



Nikola Tesla: The Eccentric

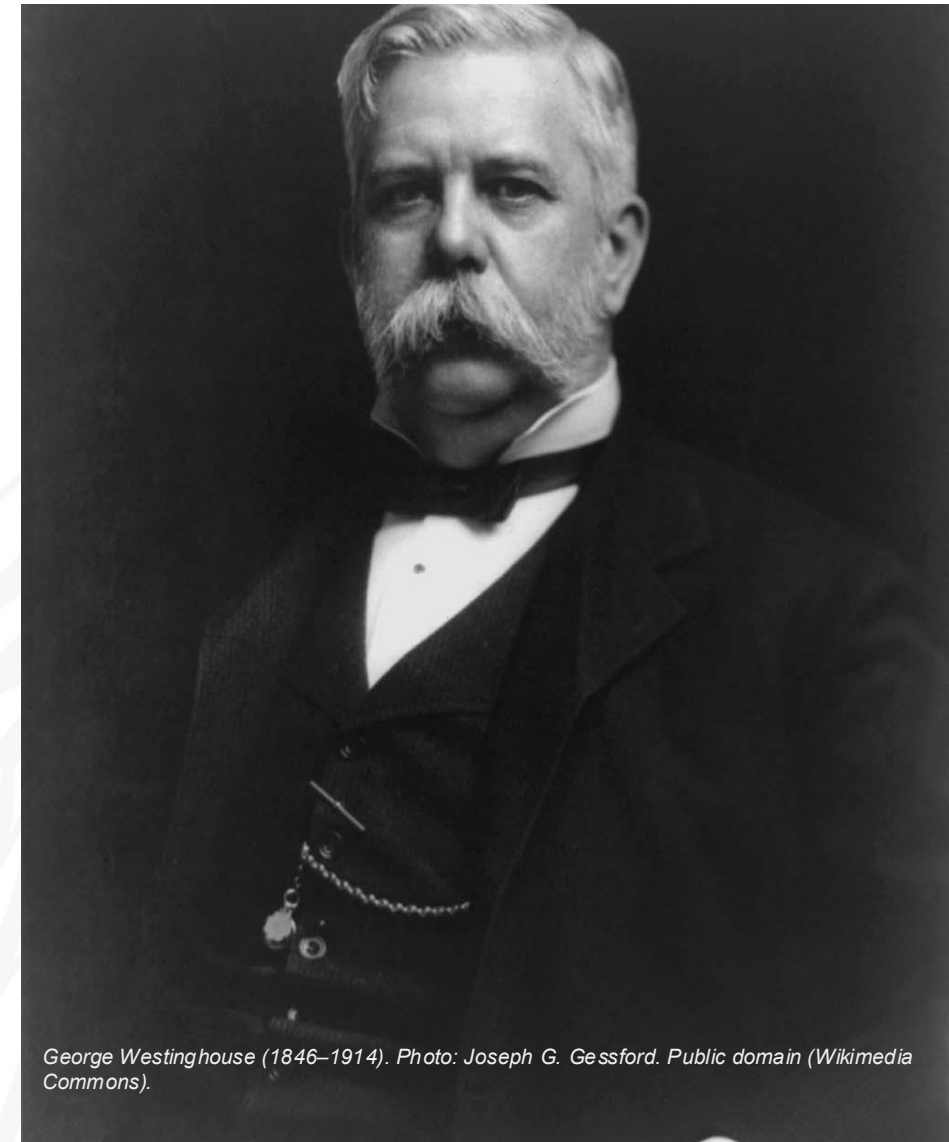
- Born in modern-day Croatia (Austria-Hungary) in 1856
- Invented the polyphase AC induction motor among many other things
 - He was not always right, believing that he could solve wireless transmission of power with EM waves, which is hugely unfeasible
- Was working for Edison, who stiffed him for bonuses repeatedly and made fun of him so he quit
- In 1888: licensed his patents to Westinghouse. Now AC had generation, transmission, AND motors: a whole system.
- Brilliant, visionary, terrible at business, and juuuustttt a liiiittllleeee extra. He died broke in a New York hotel.
 - He spent most of his money trying to care for pigeons and throwing himself lavish birthday parties where he unveiled his more... eccentric ideas



*Nikola Tesla (1856–1943), photo by Napoleon Sarony, 1890.
Public domain (Wikimedia Commons).*

George Westinghouse: The Idealist

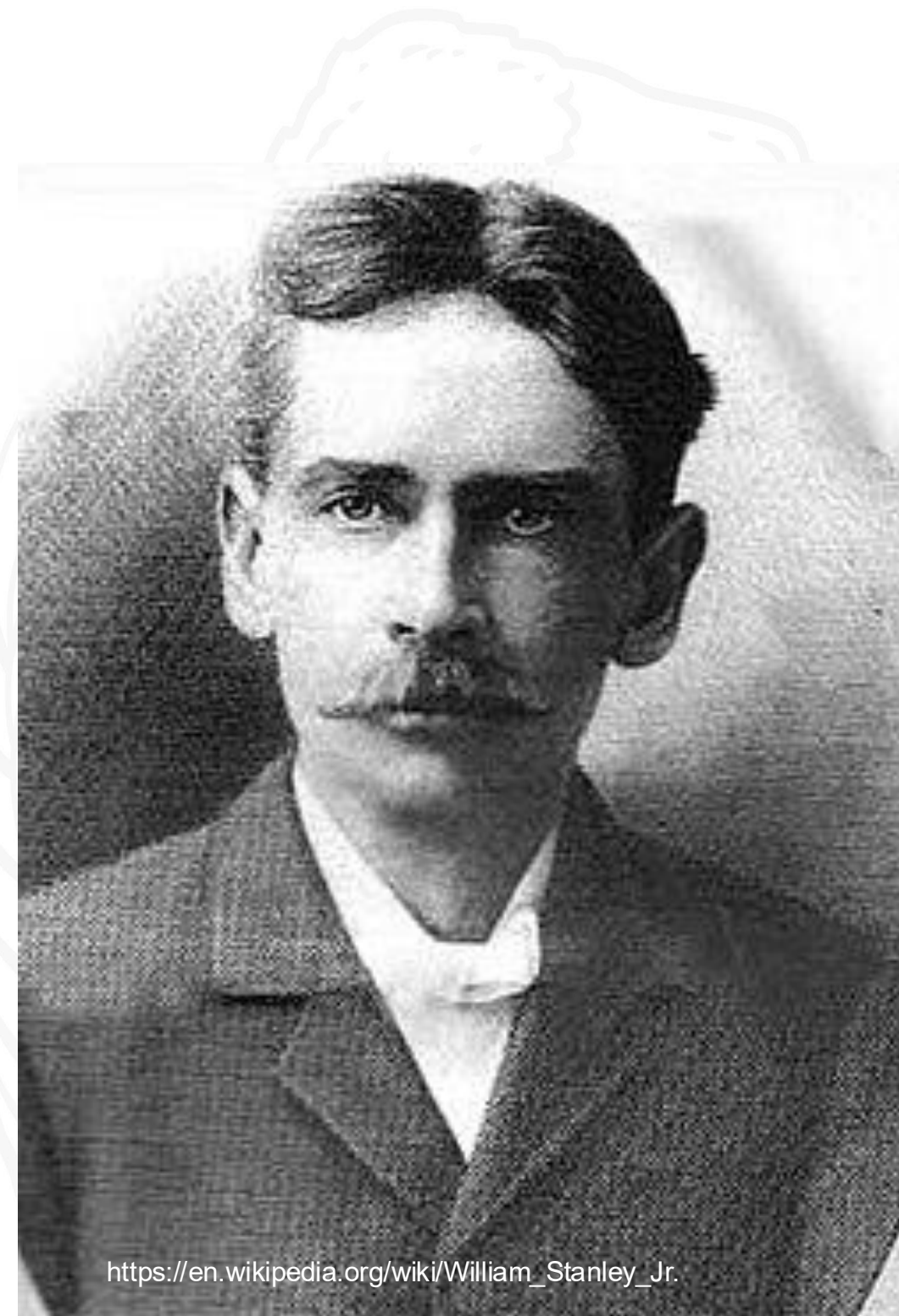
- Born in Pittsburg 1846
- Tried to run away from home at 14 to join the Union army
- Invented the railway air brake, created a way to mine natural gas, a steam turbine for boats, and was the first businessman in America to offer his employees Sundays and half of Saturdays off (5.5 day work week) as well as disability benefits, medical care
- Did experiments with AC power and saw Tesla's vision to scale it up to large power plants
- That facial hair was... well... popular at the time
- He also saw Tesla's descent into poverty due to extravagant living and had his bills taken care of until Tesla's death



George Westinghouse (1846–1914). Photo: Joseph G. Gessford. Public domain (Wikimedia Commons).

William Stanley Jr: The Transformer Guy (Not Optimus Prime)

- Born in Brooklyn in 1858
- Lived in Pittsburg, contracted tuberculosis and left due to poor air quality (well that's ironic)
- Went to work for Westinghouse and later founded his own company, Stanley Electric (subsumed into General Electric)
- Built the first practical transformer. Worked on the Great Barrington, MA (1886) and eventually the Buffalo Niagara system.



https://en.wikipedia.org/wiki/William_St Stanley_Jr.

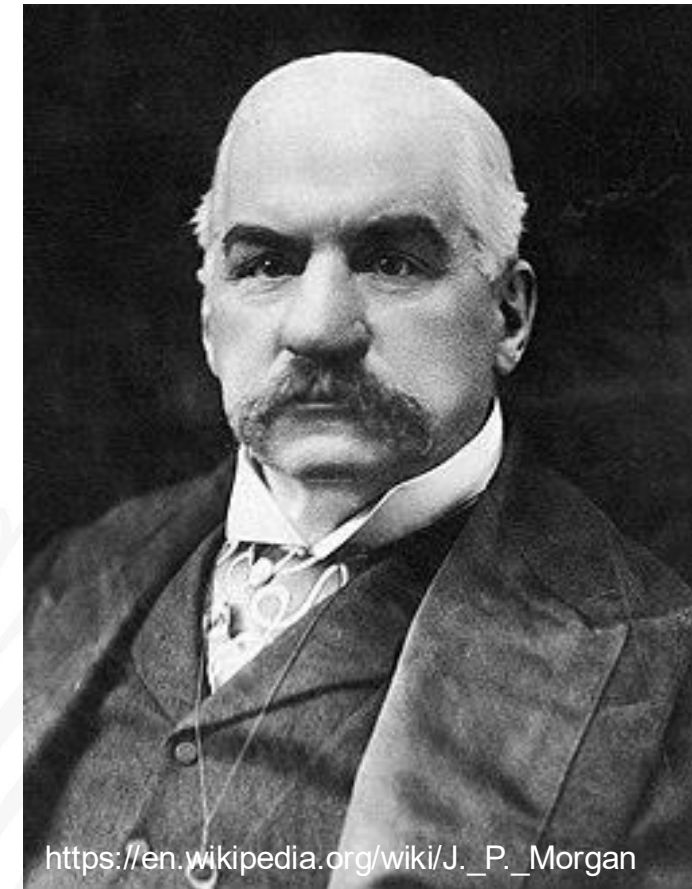
Harold P. Brown: The Rage-Baiter

- Born in Wisconsin, 1844
- Freelance engineer who ran the anti-AC campaign: public animal killings, lobbying for voltage caps, and building the state's execution equipment.
- Aug 1889: stolen letters published in the New York Sun exposed that Edison's company was funding and steering him. The “independent expert” was a plant colluding with Edison



J.P. Morgan & the Vanderbilts

- Financier J.P. Morgan (nepo baby descendent from railroad money) and the [Vanderbilt family](#) (nepo family) bankrolled Edison Electric from the start.
- Morgan engineered the 1892 merger that created General Electric and dropped Edison's name without telling him until the day before.
- The lesson: the current war wasn't settled only by better physics. It was settled in boardrooms. Finance decides which technology gets to scale.



1882: Edison Builds the First Grid (DC)

- Sept 1882: Edison's [Pearl Street Station, NYC](#), the first investor-owned electric utility. Coal-burning plant that supplied power for the arc lamps that Edison sold.
- Worked at 110 volts DC.
- Worked fine for dense Manhattan blocks but was useless for a town 10 miles away due to high losses of DC current.
- Edison had a working business and a bunch of patents to protect, so needed DC power to be adopted so he could get richer

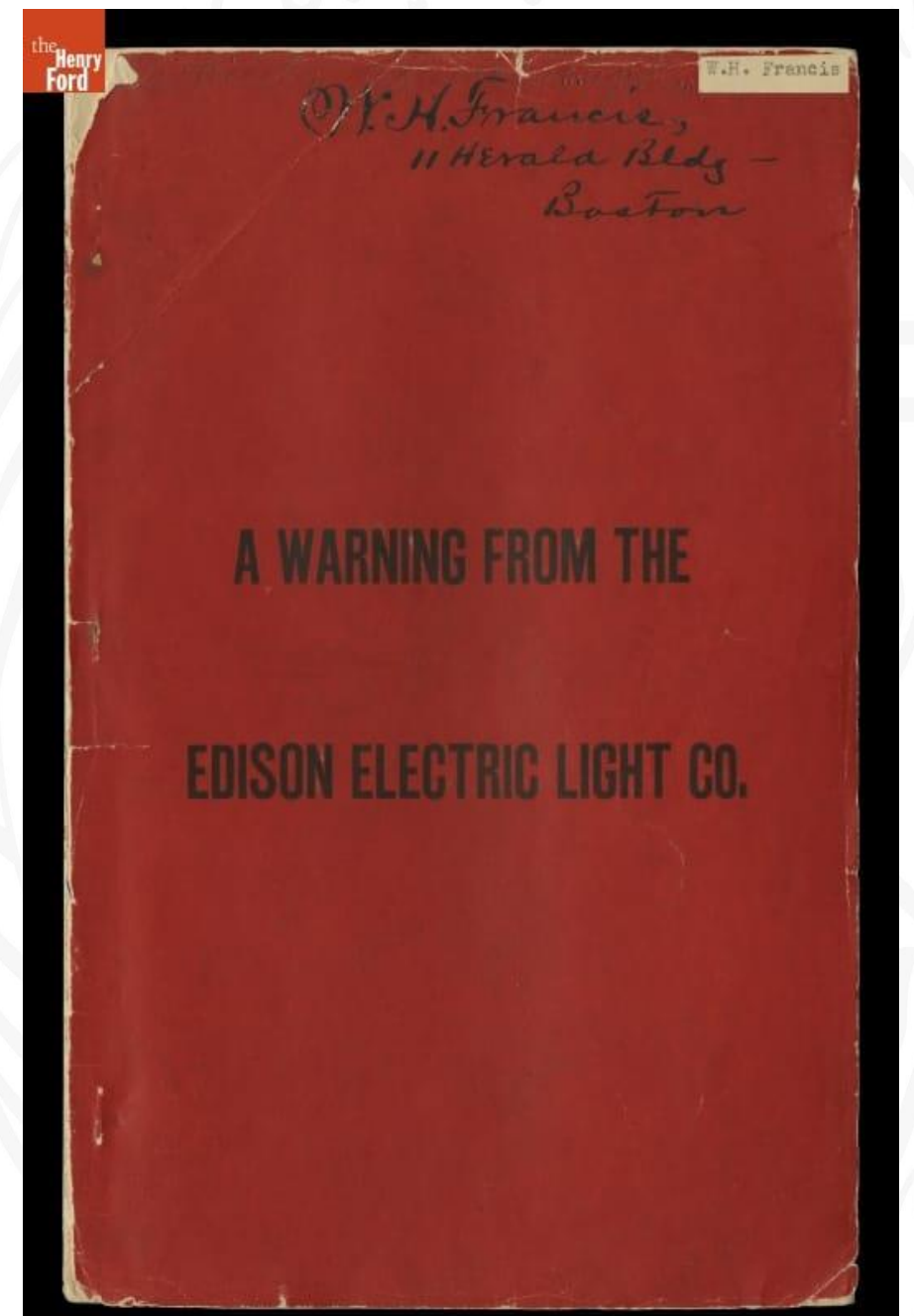


1885–1886: Westinghouse Bets on AC

- 1885: Westinghouse buys U.S. rights to the Gaulard–Gibbs transformer and hires engineer William Stanley.
- March 1886: Stanley lights Great Barrington, MA : 500 V sent down the street, stepped down to 100 V for the shops. This provided proof AC could travel far, then be made safe at the wall.
- Fall 1886: Westinghouse, Stanley & Oliver Shallenberger build the first commercial AC system right here in Buffalo, NY.
- End of 1887: Westinghouse 68 AC stations vs. Edison 121 DC. AC is catching up fast.

1886–1888: Edison's Smear Campaign

- Edison started realizing that AC was winning the current war at the market due to the transformer step up / step down combination
- Nov 1886, Edison said privately: "Just as certain as death, Westinghouse will kill a customer within six months".
- Feb 1888: Edison Electric publishes an 84-page pamphlet, "A Warning": part patent threat, part fear campaign cataloguing AC deaths.
- If you can't beat AC on cost or physics, brand it as the current that kills! **19th-century corporate disinformation campaign.**



1888: Safety Concerns Mount

- Due to lack of regulation, power lines EXPLODED over NYC without standards for safety
- Blizzard of 88 downed power lines, leading to the death of a child
- People tried to force the power companies to put their wires underground, but a court battle (funded by the companies) delayed the implementation

This is a critical example of a (useful!) technology being deployed without proper regulation that leads to deaths and danger to the population... gee I wonder if we've learned our lesson 140 years later???



1888: AC power KILLSZSZZ!!11!

- June 1888: Instead of adding regulations about the actual problem (unsafe wires being put up in haste), engineer Harold P. Brown attacks AC in the New York Post and lobbies NYC to cap AC at 300 V, which would make long-distance AC useless.
- However, secretly bankrolled and equipped by Edison's company: treasurer Francis Hastings brokered it and Edison lent his West Orange lab and his assistant Arthur Kennelly.
- Brown stages public animal electrocutions as “proof” AC is deadlier than DC (see the electric-chair section).

Paid children to capture stray dogs so he could execute them in the streets of NYC



Modern equivalent would be a social content creator that manufactures outrage by rage-baiting people for money...

1889: The Chair, The Panic, and the birth of General Electric

- May 1889: William Kemmler sentenced to die in the new electric chair (details ahead).
- Oct 11, 1889: lineman John Feeks is electrocuted in a tangle of Manhattan wires in front of a crowd, sparking the “Electric Wire Panic.”
- Jan 1889: Edison General Electric is formed, backed by J.P. Morgan and the Vanderbilts and Edison starts losing control of his own company.
- Changed name to General Electric without Edison’s consent or knowledge



1890–1891: AC Wins the Market

- Aug 6, 1890: Kemmler executed by AC — a botched, grisly spectacle meant to smear AC. It didn't stop AC's spread.
- 1891: Dolivo-Dobrovolsky demonstrates three-phase AC, 176 km transmitted at ~75% efficiency. The long-distance problem is now solved. DC has no answer.
- Even Edison's own Machine Works quietly starts building AC gear (Oct 1890).



1892–1893: AC Wins the Current War

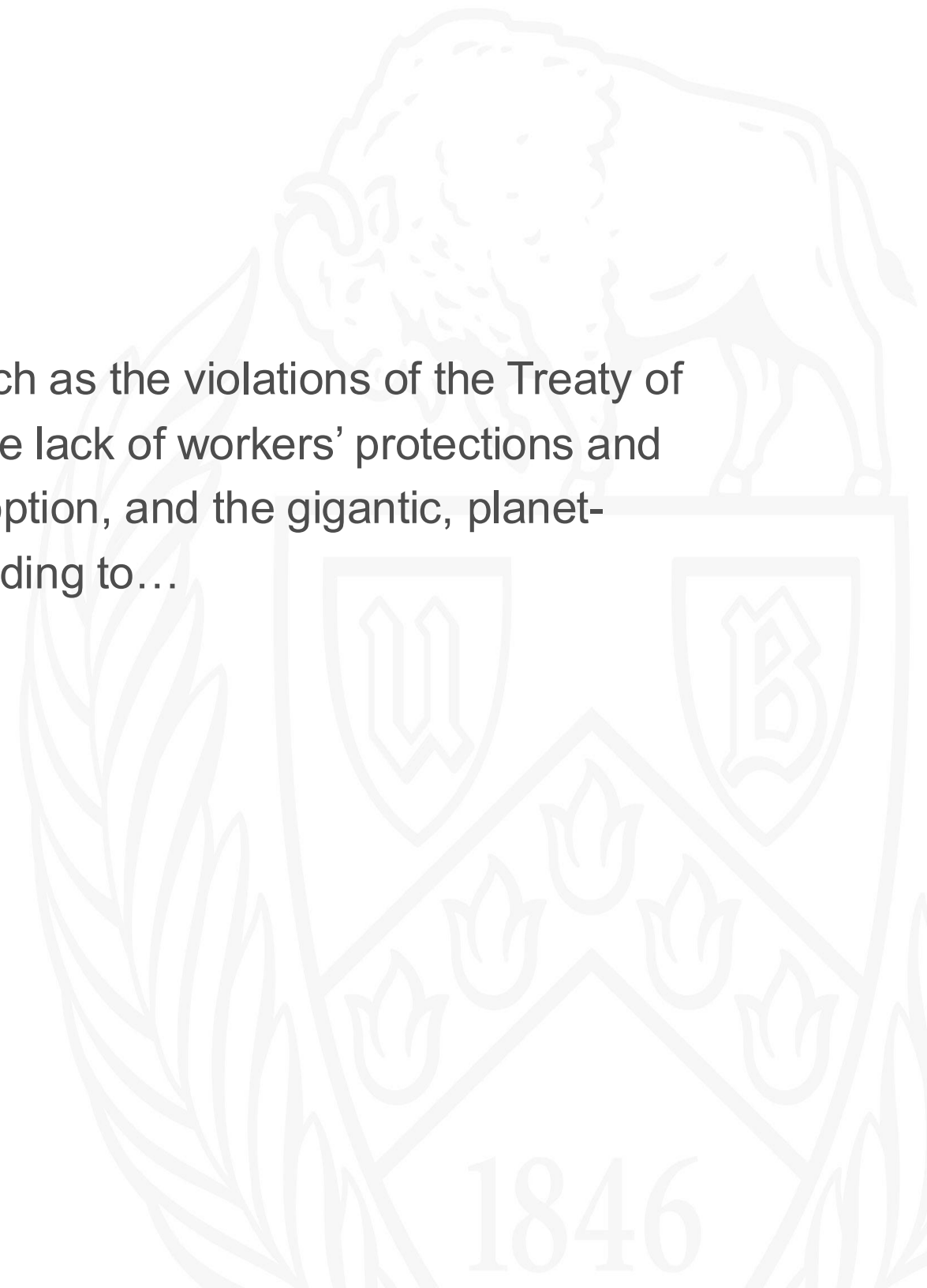
- April 1892: Edison General Electric + Thomson-Houston merge into General Electric — Edison's name is dropped.
- 1893: Westinghouse safely lights the [Chicago World's Fair](#) with AC
- 1893: Westinghouse wins the Niagara Falls hydro contract and GE builds the transmission lines. [Adams Power Plant](#) built. By 1896 AC power flows from Niagara to Buffalo, 20+ miles away. First major large-scale power plant in human history. It revolutionized humanity and showed how to power an entire country.
- This ended the War of the Currents



Everyone switched to AC power
They all got rich
THE END

To come: The human cost

- We've touched on the human cost a little already
- There are a lot of stories we have not yet unpacked, such as the violations of the Treaty of Canandaigua (again) to build the Niagara Falls plant, the lack of workers' protections and safety protocols for power lines despite widespread adoption, and the gigantic, planet-altering impact of burning coal for almost 200 years, leading to...



Discussion from the end of last class

#	Topic cluster	Students	Individual comments
1	Artificial Intelligence	9	"New AI and how much it should be used"; "AI"; "AI"; "The use of AI vs not using AI"; "AI with all the hype and funding that goes into it"; "many different AI software and companies competing... for money"; "AI data centers and the demand on power versus humanity"; "Data centers"; "Copyright of AI generated content"
2	Electric vs. gas cars / EV industry	5	"Electric Car vs Gasoline Car"; "Gas car vs electric car"; "Tesla vs every other car brand"; "Vinfast in Vietnam against... any other electric car company"; "The switch of fuel in Vietnam"
3	Consumer tech & device brands	5	"Apple vs Samsung"; "Apple vs Samsung"; "Apple vs Android"; "Alexa vs Google"; "Game Companies like Sony and Microsoft"
4	Energy generation & fuels	2	"Solar vs Nuclear"; "Fracking"
5	Batteries / energy storage	1	"Lithium Batteries"
6	E-commerce & shipping	1	"Different types of mail companies like Amazon"
7	Surveillance tech	1	"Flock cameras"
8	Social / political	1	"Pro-choice vs pro-life"

Discussion from the end of last class



Claude's Guess at the issue

Give me a 1-2 sentence summary of what people think are the key issues in each of these areas.

- **Artificial Intelligence** — This is the dominant concern, and it's really several worries bundled together: how much AI should be used and trusted at all, the hype and money pouring into competing AI companies, the enormous power and data-center demands AI places on the grid "versus humanity," and who owns the copyright to AI-generated content. The through-line is unease about a technology scaling faster than its costs and rules are understood.
- **Electric vs. gas cars / EV industry** — Students are focused on the competition between electric and gasoline vehicles and, within EVs, the race among manufacturers (Tesla versus everyone, and VinFast versus rivals). Vietnam comes up repeatedly as a live example of a national "switch of fuel" from gas to electric.
- **Consumer tech & device brands** — The interest here is head-to-head brand rivalry — Apple vs. Samsung, Apple vs. Android, Alexa vs. Google, Sony vs. Microsoft — essentially who wins the battle for dominance in phones, voice assistants, and gaming.
- **Energy generation & fuels** — A smaller group is weighing how we should produce energy, framed as solar versus nuclear and the pros and cons of fracking. The key issue is which energy sources we should rely on going forward.
- **Batteries / energy storage** — One student flags lithium batteries, pointing at the storage technology (and its supply/materials questions) underpinning both EVs and renewable energy.
- **E-commerce & shipping** — One response raises competition among shipping and delivery companies like Amazon, i.e., who controls how goods get to people.
- **Surveillance tech** — One student names Flock cameras, signaling concern about automated license-plate readers and everyday surveillance.
- **Social / political** — One response raises the pro-choice vs. pro-life debate, the lone purely social-issue topic in the set.

Claude's Guess at the controversies

Give me a 1-2 sentence summary of the core controversies in each of these topics.

- Artificial Intelligence — The core disputes are whether AI's rapid deployment is outpacing safety, accountability, and job protections, and whether its massive energy and water demands are justified; a parallel legal fight rages over whether training on and generating from copyrighted work is fair use or theft, with courts and regulators still unsettled.
- Electric vs. gas cars / EV industry — The debate is whether EVs are genuinely cleaner once battery mining, manufacturing, and grid electricity are counted, and whether they can scale given charging infrastructure, cost, and range concerns. There's also a competitive/geopolitical fight over subsidies and market dominance, with newer players like China's and Vietnam's automakers challenging Tesla and legacy brands.
- Consumer tech & device brands — Beyond brand loyalty, the real controversies are over walled ecosystems and lock-in (Apple's App Store and repairability fights), antitrust scrutiny of a few dominant firms, and privacy — how much always-listening assistants and connected devices surveil their users.
- Energy generation & fuels — Solar vs. nuclear pits intermittency and land use against nuclear's cost, waste, and safety fears, while fracking is contested over groundwater contamination, methane leakage, and induced earthquakes versus its role in cheap energy and independence. The overarching dispute is how fast to transition off fossil fuels and what the reliable replacement is.
- Batteries / energy storage — The controversies center on the environmental and human cost of mining lithium, cobalt, and nickel (often in conditions raising labor and geopolitical concerns), plus recycling, fire safety, and whether supply can meet demand without new bottlenecks.
- E-commerce & shipping — The friction is over the market power of giants like Amazon — their effect on small retailers and warehouse/delivery labor conditions — alongside the environmental toll of fast, packaging-heavy shipping.
- Surveillance tech — Flock and similar automated license-plate-reader networks raise the core tension between public-safety/crime-solving benefits and mass warrantless tracking, data retention, and the lack of consent or oversight over who accesses the data.
- Social / political — The abortion debate turns on the fundamental, and largely irreconcilable, disagreement over when life and personhood begin and how to weigh fetal rights against bodily autonomy — a moral and legal question that people of good faith answer very differently.

Claude's Guess at your answers

Now, the prompt to the students making these responses was "What are some scenarios in the 21st century that have a parallel with the War of the Currents?" Give a guess as to what the students meant by their responses.

Claude's Guess at your answers

- Good context — that reframes the whole set. The War of the Currents was a standards war over a transformative general-purpose technology: two incompatible systems (Edison's DC vs. Westinghouse/Tesla's AC) fought to become the dominant infrastructure, driven by corporate rivalry, huge capital, network effects, and public fear campaigns — with the technically superior standard ultimately winning. Read through that lens, here's what students likely meant:

Claude's Guess at your answers

- Artificial Intelligence — Most of these students probably see the AI-company race (OpenAI, Google, Anthropic, etc.) as a modern standards war: rival firms burning enormous capital and hype to become the dominant platform everyone builds on, winner-take-most. The "data centers / power demand" responses are an especially sharp parallel — they've noticed the fight has literally come back to electricity, with AI's grid and energy strain echoing the infrastructure buildout the original currents war was about.

9 Responses

Claude's Guess at your answers

- Electric vs. gas cars / EV industry — This is one of the cleanest intended parallels: electric vs. internal-combustion as two competing propulsion "standards," with charging networks vs. gas stations mirroring AC vs. DC grid buildout, and Tesla/VinFast/legacy automakers as the rival "Edison and Westinghouse." (There's even a literal modern version they may or may not know about — Tesla's NACS plug vs. CCS as the EV-charging standard war.)

5 Responses

Claude's Guess at your answers

- Consumer tech & device brands — These are textbook format/platform wars, the most direct heirs to the analogy: iOS vs. Android, Apple vs. Samsung, Alexa vs. Google, Sony vs. Microsoft — incompatible ecosystems competing to lock in users and become the default, exactly like AC vs. DC (and its descendant, VHS vs. Betamax).

5 Responses

Claude's Guess at your answers

- Energy generation & fuels — Solar vs. nuclear (and fracking) is almost the same fight one century later: which technology should power the future, decided by cost, safety fears, and infrastructure lock-in. The student sees rival energy "standards" competing for dominance, with public perception and safety controversy as a deciding factor — very much the shape of the original.
- Batteries / energy storage — Likely meant as the enabling infrastructure whose "standard" is being fought over (lithium and competing chemistries), the way the grid was the substrate of the AC/DC battle — plus a competition over who controls the critical technology everything else depends on.

2, 1 Responses

Claude's Guess at your answers

- E-commerce & shipping — The student probably reads Amazon vs. other delivery/logistics companies as a battle to own the dominant distribution network, where network effects and scale let one winner entrench — the infrastructure-monopoly dimension of the currents war more than the technology dimension.
- Surveillance tech (Flock cameras) — This is more of a stretch, and the parallel they intended is probably the public-fear/safety angle: a powerful new technology being rapidly deployed as infrastructure before society agrees it's safe or acceptable — echoing the electrocution-scare PR battles that surrounded the rollout of electric power.

1 Response each

Claude's Guess at your answers

- Social / political (pro-choice vs. pro-life) — This one doesn't really fit the standards-war framing at all; that student most likely read the prompt loosely as "any two-sided modern conflict" rather than a competing-technology/infrastructure battle.

1 Response

Claude's Guess at your answers

- One takeaway worth noting for your teaching: the strongest answers grabbed the standards-war-over-a-transformative-technology structure (EVs, iOS/Android, solar/nuclear, AI platforms), while a few — pro-choice/pro-life, arguably Flock cameras and the AI-copyright response — drifted toward "any rivalry or controversy." That split might be a useful thing to surface back to the class when you debrief what the War of the Currents analogy is actually pointing at.

More than half of these are directly or peripherally related to climate change

Discussion

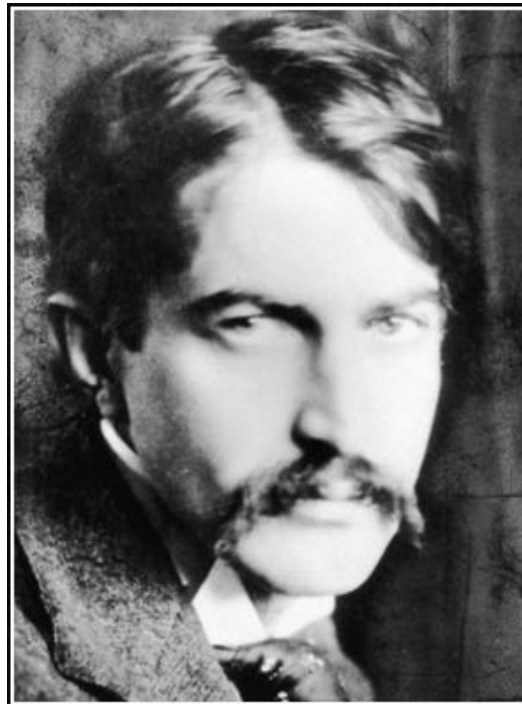
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ANTHROPOGENIC CLIMATE CHANGE



Scientific Debate

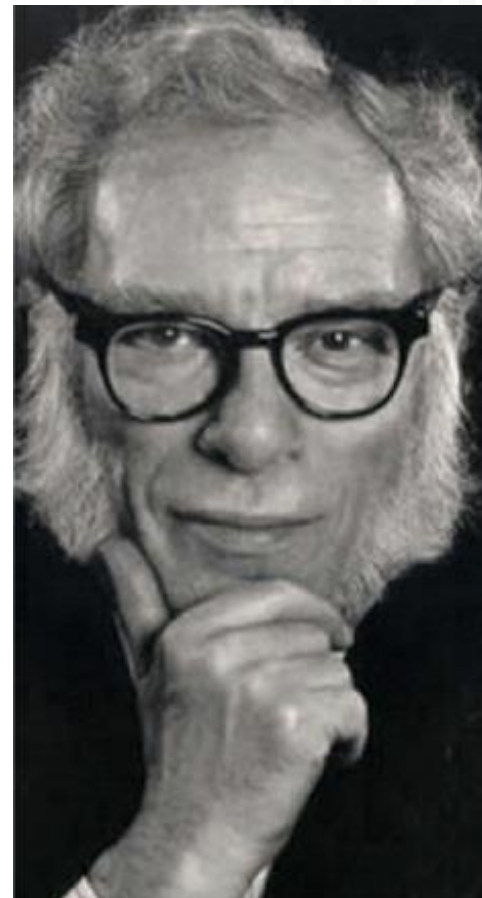
- Fact based, supported with evidence
- Actually NOT a democratic process
 - Nature does not care about our opinions
- Also not post-modernistic
 - There IS a right answer and a wrong answer
 - Social controversy is NOT the same as scientific debate



A man said to the universe: 'Sir, I exist!' 'However,' replied the universe. 'The fact has not created in me A sense of obligation.

— Stephen Crane —

AZ QUOTES

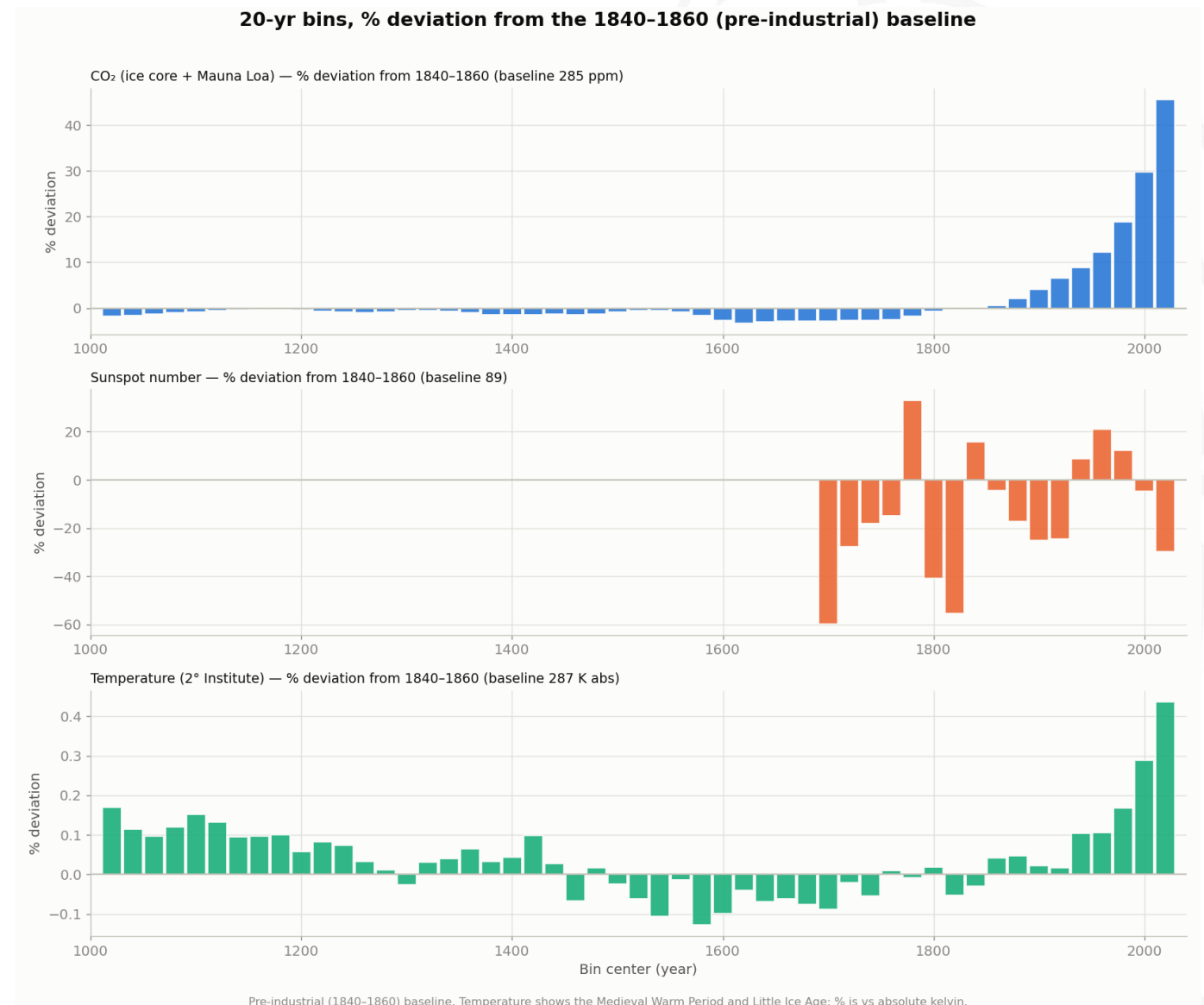


"Anti-intellectualism has been a constant thread winding its way through our political and cultural life, nurtured by the false notion that democracy means that 'my ignorance is just as good as your knowledge.'"

Isaac Asimov

Anthropogenic climate change

- Burning fossil fuels releases **CO₂**, which traps heat
- The scientific verdict is **unequivocal**: human activity (anthropogenic) has warmed the planet
- A lot of non-scientific outlets like to point to “sun weather” (i.e. sun spots) as the cause for temperature variations
 - But how can the temperature go UP since 1950 when the number of sunspots has gone DOWN?
- The fossil fuel industry has been misrepresenting this for decades because they have a trillion-dollar stake in you purchasing fossil fuels.

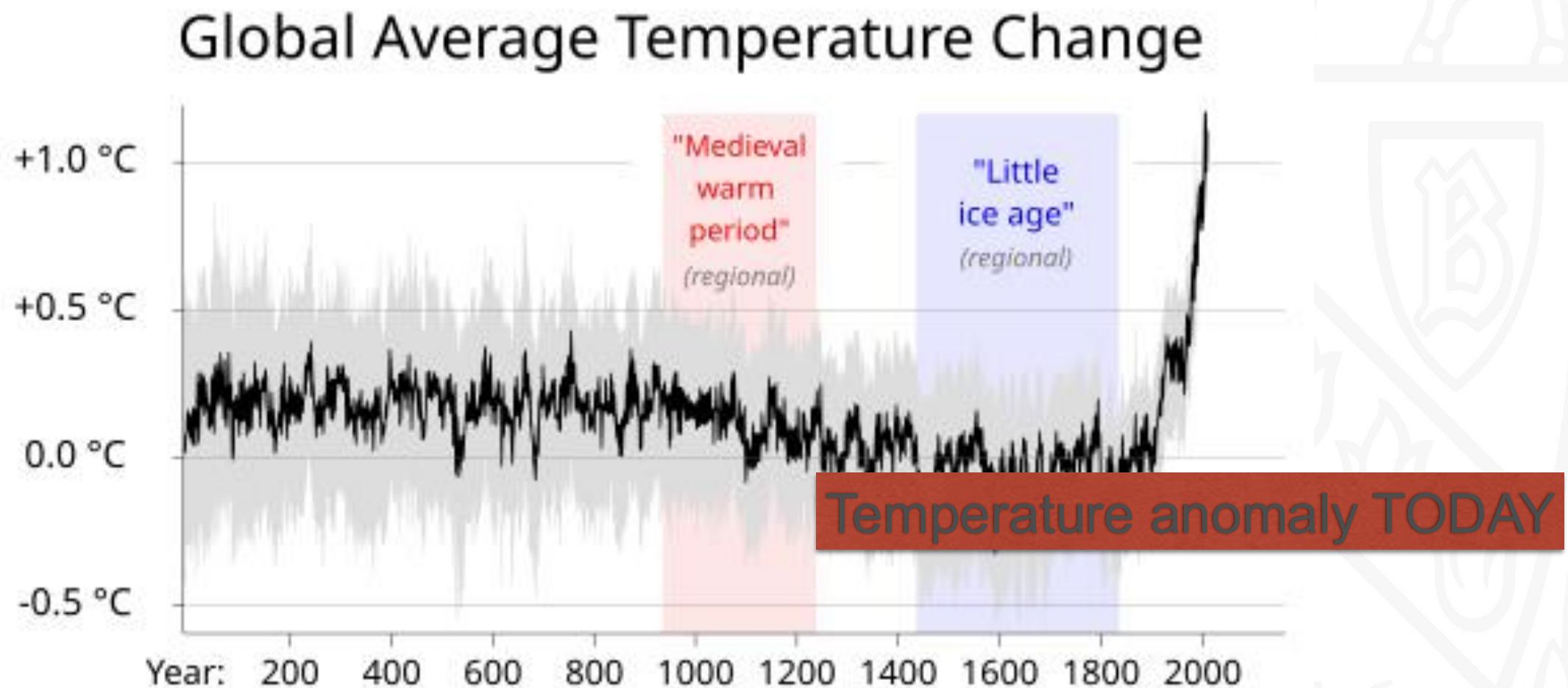


Scientific Debate

- Examples of controversial topics that are NOT in scientific question whatsoever:
 - **Evolution** (solved in the 1860's)
 - **Age of the earth and universe** (solved in the 1910's)
 - **Anthropogenic climate change** (solved throughout the latter part of the 20th century, and evidence grows stronger)
 - **Nuclear power** (statistically one of the safest forms of power on earth!)
 - **Vaccination** (does not cause autism)
 - **Perpetual motion** (read Faraday's law!)
- Examples of topics that ARE up for scientific debate :
 - What is the nature of **gravity** at small scales?
 - Is there **life on other planets**?
 - Why is there **more matter than antimatter**?
 - Can we model **turbulence** exactly? ([Navier-Stokes hoopla ongoing!](#))

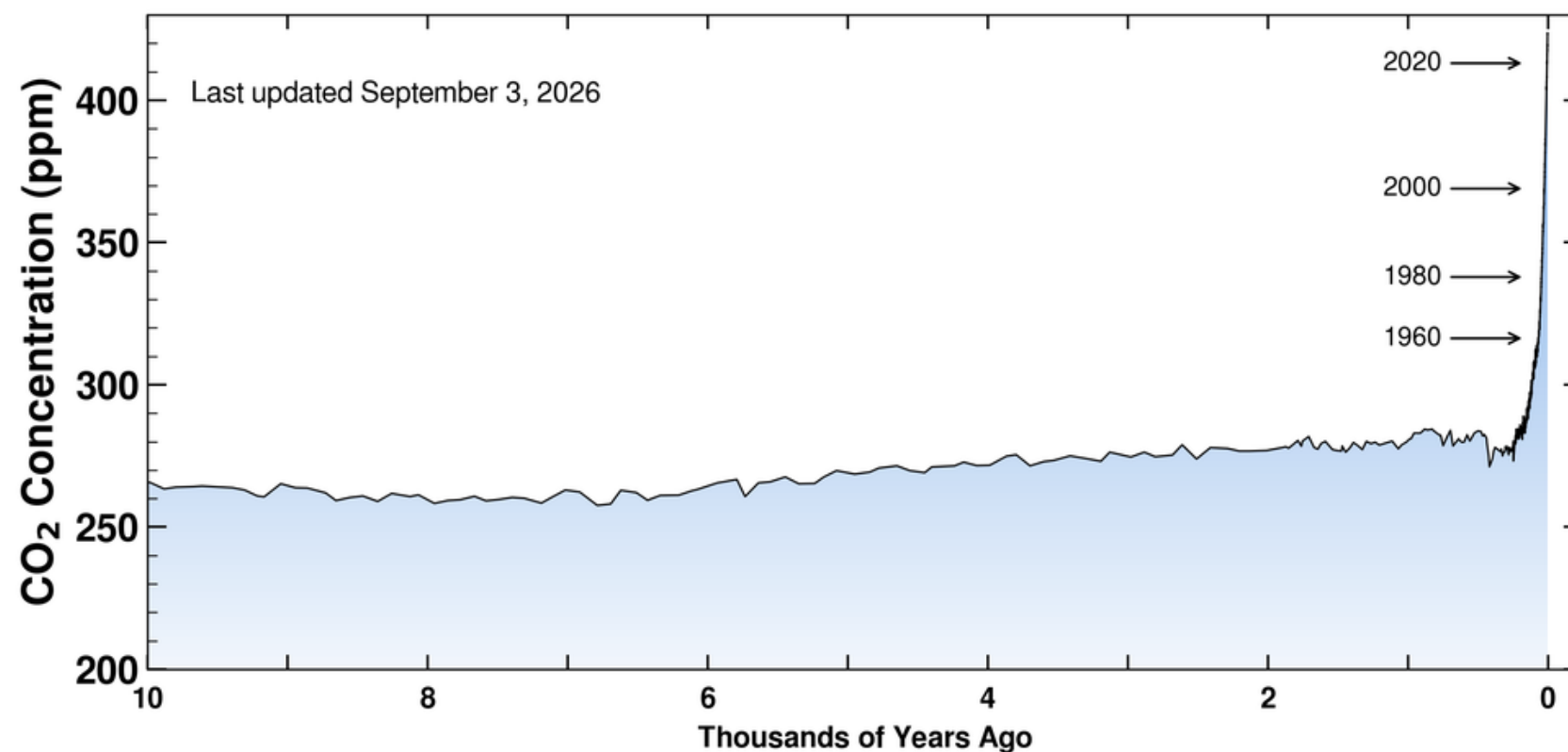
Climate Change : NOT a scientific debate

- Temperature of the earth is hotter than it has been in over 1000-2000 years, and is much larger than natural variations



Climate Change : NOT a scientific debate

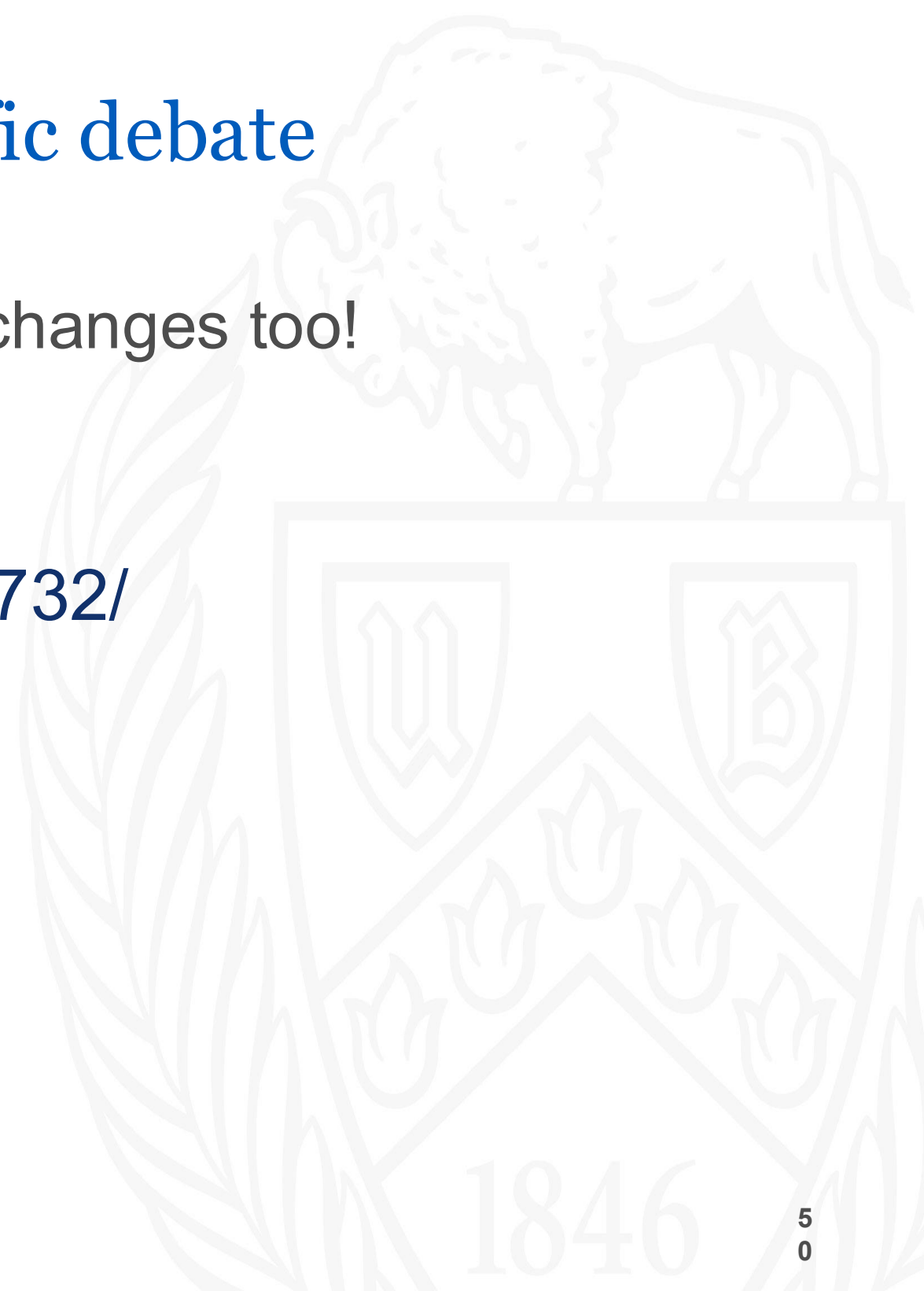
- CO₂ is rising quickly
 - Extrapolating the current trend will make an atmosphere that humans literally cannot breathe due to a toxic level of CO₂ in ~1000 years
 - Other damage will occur far before that



Climate Change : NOT a scientific debate

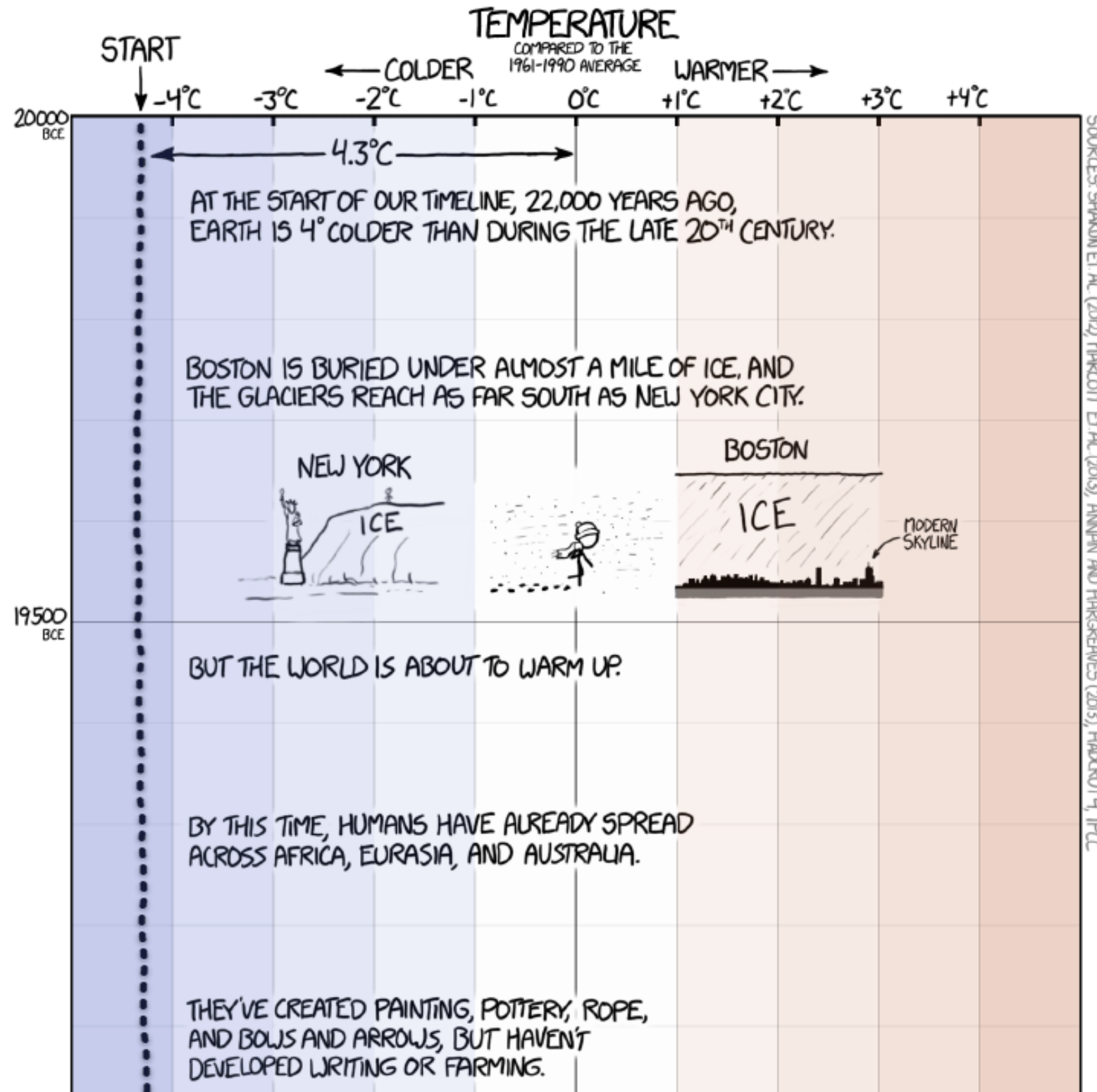
- But natural variations cause climate changes too!
 - Not this much, and not this quickly.

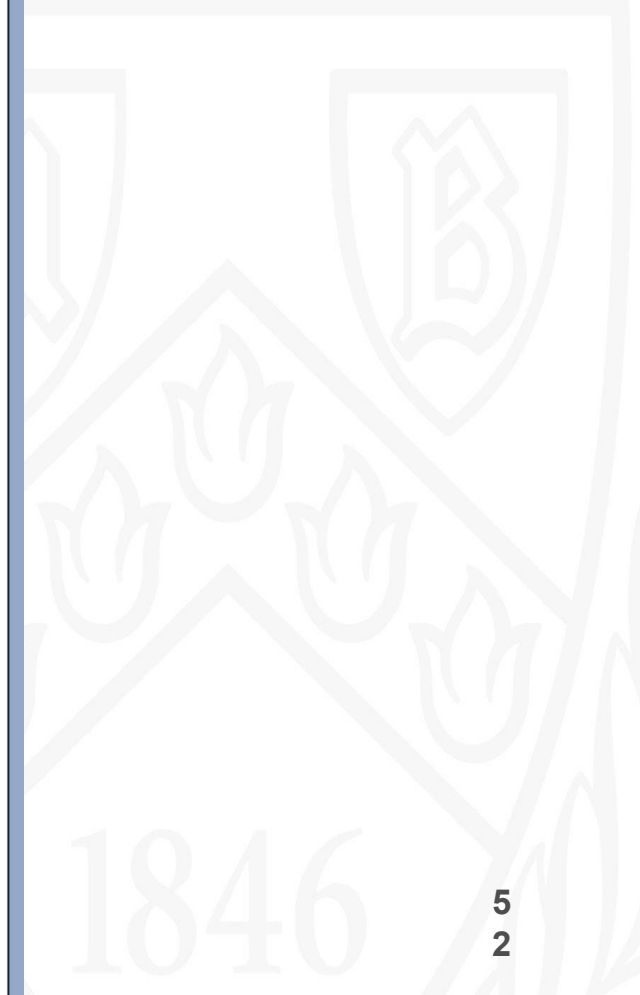
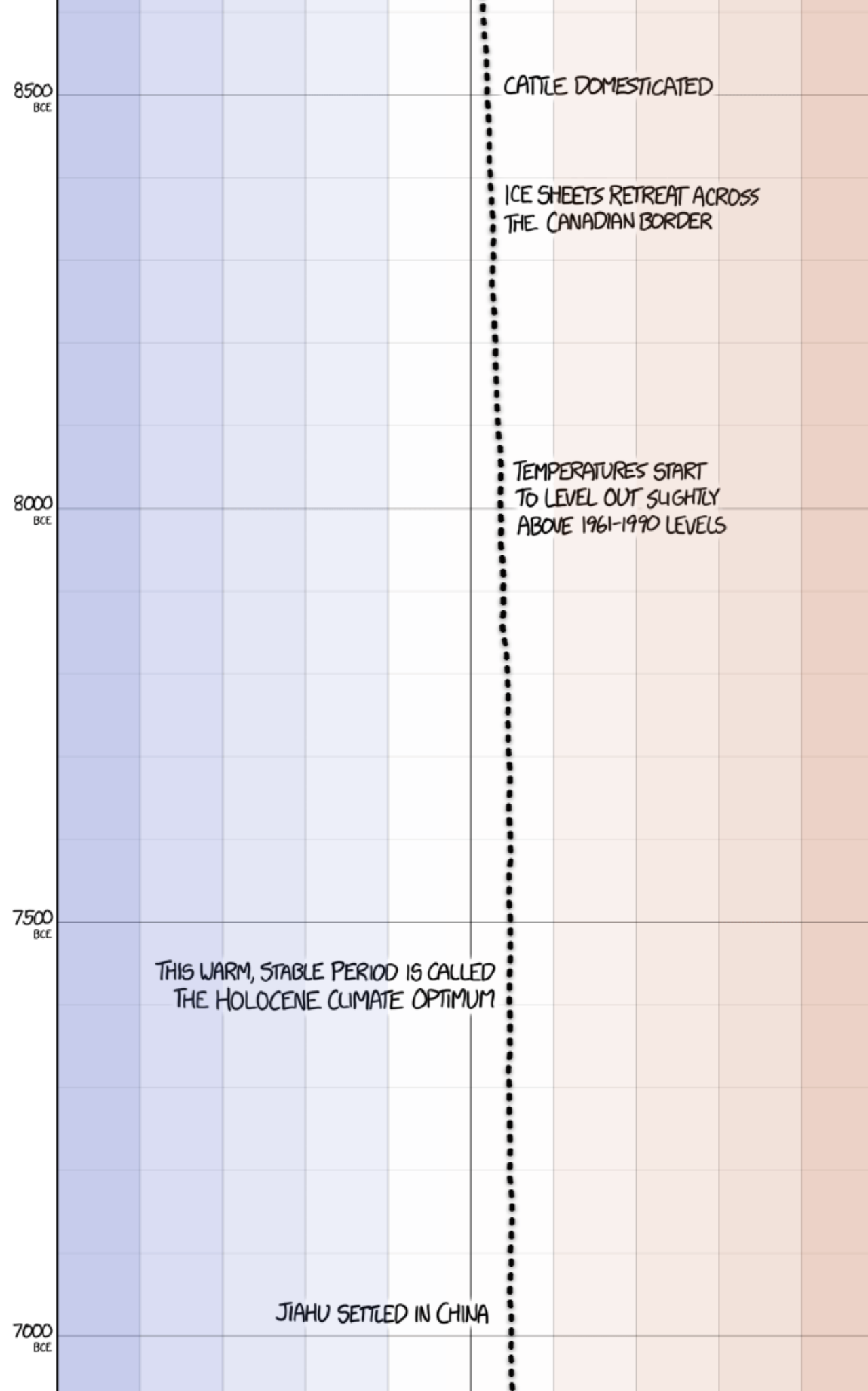
<https://xkcd.com/1732/>

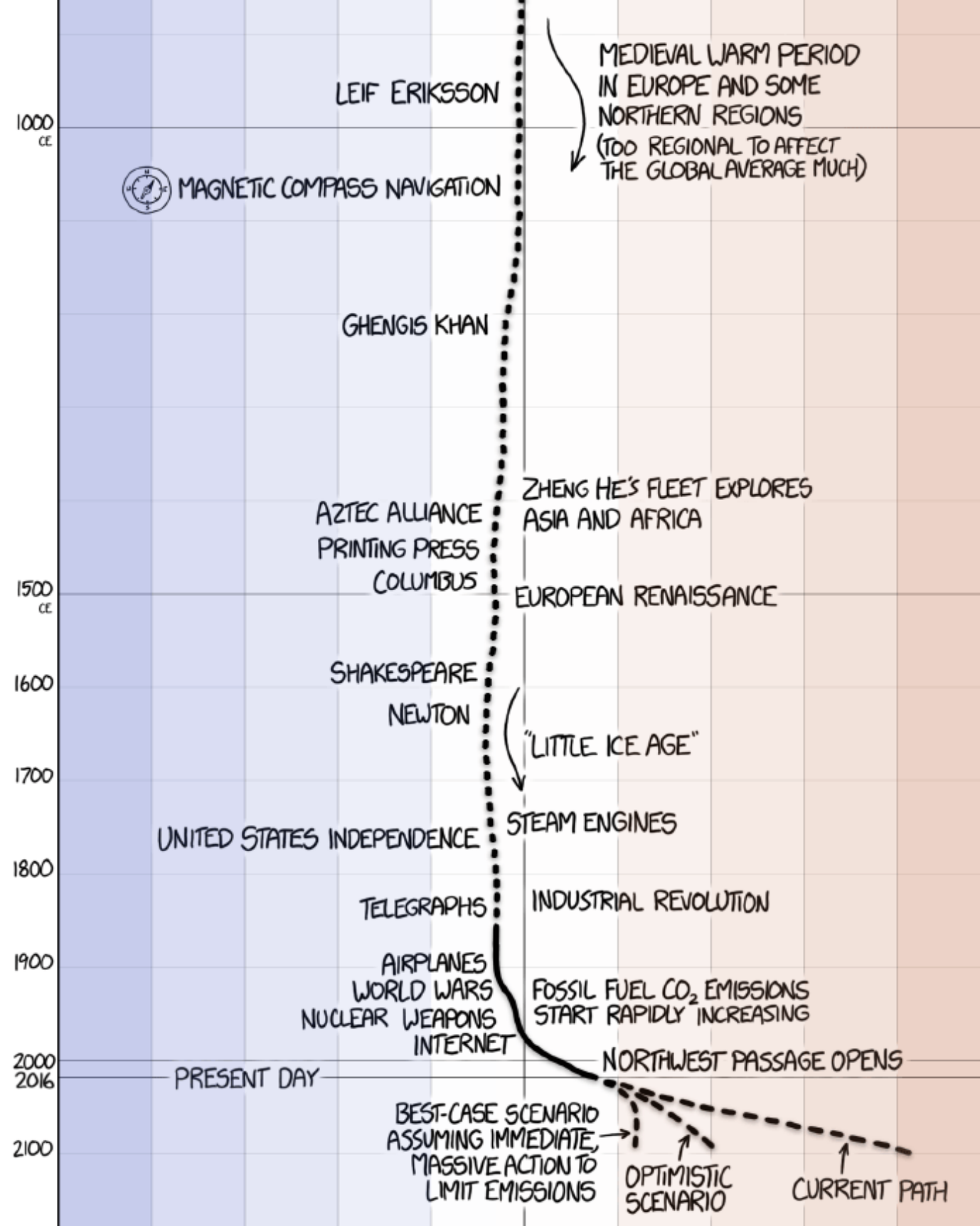


A TIMELINE OF EARTH'S AVERAGE TEMPERATURE SINCE THE LAST ICE AGE GLACIATION

WHEN PEOPLE SAY "THE CLIMATE HAS CHANGED BEFORE," THESE ARE THE KINDS OF CHANGES THEY'RE TALKING ABOUT.





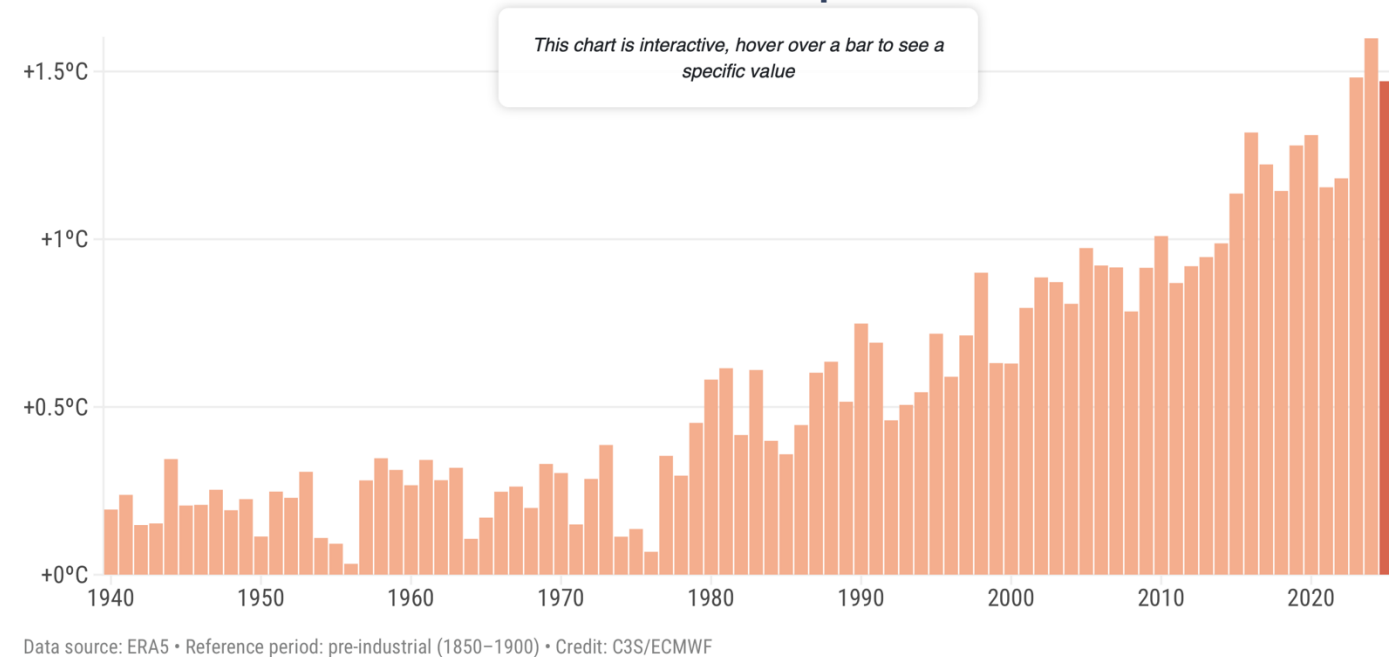


Climate Change : NOT a scientific debate

- But so-and-so doesn't believe in global warming!
 - Doesn't matter. Not a democracy. Data wins.
- But CO2 lags temperature in the ice cores!
 - Actually recently solved! Bubble propagation in ice can take several hundred years (Science 01 Mar 2013:Vol. 339, Issue 6123, pp. 1060-1063)
- But it hasn't gotten hotter since 1997!
 - Yes, it actually has. This myth was started in 2005.
 - The hottest years in human history were:
 - 2024
 - 2023 & 2025 (tie)
 - The hottest 11 years **ever** were the previous 11 years
- Several of my graphics were not updated in 10 years or so, the last taught something like this.
 - Guess what? The messages were exactly the same.
 - Do this again in 10 years, and it will STILL be the same.



Global annual surface air temperature anomalies



<https://climate.copernicus.eu/global-climate-highlights-2025>

Climate Change : NOT a scientific debate

- But I hate winter! If it gets warmer, we won't have winter, right?
 - We'll still have winter. It may even cause more extreme temperature swings to make them WORSE
- But the scientific conspiracy! Well-educated social elites and foreigners are lying to the common people for self interest!
 - (Of course, it isn't true... otherwise could someone tell me how to collect my check for being part of this big money conspiracy?
Is there a form to fill out, orrrrr....?)
 - Sounds familiar though, right?

Climate Change : NOT a scientific debate

- **Why so contentious in the political realm?**

- First : It shouldn't be. Tell your Representatives and Senators that, regardless of party. Science is not a partisan issue.
- Second : Money.
 - Coal and oil used to employ a lot of people
 - Automation replacing humans in factories and mining (Information Age economic revolution)
 - Emergent new technologies are competing with existing technologies
 - Fact of the matter is, we aren't going back to the coal-and-steel days any more than the hand-woven textile days

References & Sources

- War of the currents — overview: Wikipedia, used as a finding aid (accessed Sep 2026), traced to the sources below.
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- Images: Wikimedia Commons — Edison (Bachrach, PD), Westinghouse (Gessford, PD), Tesla (Sarony, PD), electric chair (PD). All public domain.
- AI-usage disclosure: drafting/sourcing assisted by Anthropic Claude (Opus-class), from the cited books via Wikipedia. Anthropic data centers: primarily US. Est. compute: ~minutes of GPU time (Sal to refine per course AI-cost policy).